

MUTONGA ROMEO

Software Engineer

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[LinkedIn](#) | [Work Profile](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software Engineer with 2+ years building, testing and maintaining production web and mobile applications. Works in TypeScript and React with React Native for mobile on the front end, and Node.js, Express and PostgreSQL on the back end. Practical experience with mobile money payment integration, escrow and milestone-based disbursement, transaction reporting, authentication and role-based access control, and the numerical accuracy that financial interfaces depend on. Delivered through automated testing, code review and Git pull request workflows on a live platform serving 1,700+ verified providers across Kenya.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3, Java, Python

Frontend: React 18, React Native (Expo), Vite, Tailwind CSS, Radix UI primitives, reusable component libraries, TanStack React Query, React Hook Form, Zod validation, React Router, Recharts, list virtualisation, responsive and mobile-first layouts

Backend and APIs: Node.js, Express, Prisma ORM, RESTful API design and consumption, Swagger/OpenAPI documentation, Socket.IO, scheduled jobs, AWS S3 presigned uploads, third-party integration (M-Pesa, Firebase, Nexus Wave AI)

Databases: PostgreSQL, MongoDB, relational schema design, SQL queries

Web security: JWT and JWKS, OAuth2 and OpenID Connect sign-in, session management, bcrypt password hashing, TOTP two-factor authentication, role-based access control, Helmet, CORS, input validation, XSS sanitisation

Testing and debugging: Vitest, Jest, Node test runner, unit, integration and smoke tests, API contract checks, browser DevTools, root-cause investigation

Tools and practice: Git, GitHub, feature branching, pull request review, Agile/Scrum, ESLint, Prettier, Linux and bash tooling, technical documentation

PROFESSIONAL EXPERIENCE

Software Developer (Contract) | JaGedo Innovations Limited — Nairobi, Kenya | Feb 2025 – Present

- Build and maintain JaGedo (jagedo.co.ke), a live construction marketplace connecting customers with 1,700+ verified fundis, professionals, contractors and hardware suppliers across all 47 counties, spanning a React web app, a React Native mobile app and a Node.js API.
- Build both the React web application and the React Native mobile application, on a reusable component library that cut duplicate UI code and shortened delivery of new screens by approximately 40%.
- Manage server state with TanStack React Query and form validation with React Hook Form and Zod schemas; virtualise long result lists to keep rendering responsive as data volumes grow.
- Design, build and consume documented REST APIs in Node.js, Express and TypeScript over PostgreSQL with Prisma ORM, published through Swagger/OpenAPI, covering authentication, job requisition, bidding, evaluation, awarding, payments and order fulfilment.
- Implement authentication and authorization: JWT with JWKS verification, session handling, Google and LinkedIn OAuth2/OpenID Connect sign-in, bcrypt password hashing, TOTP two-factor authentication, and role-based access control across five distinct user types.
- Apply web security practices including Helmet, CORS policy, server-side input validation, XSS sanitisation, and an automated build check that blocks secrets from being bundled into client output.
- Build the admin reporting and analytics module: dashboards over users, jobs, orders and products with lifecycle breakdowns, trend charts and configurable date ranges, with CSV, Excel and PDF export; maintain a shared currency and date formatting layer with fixed decimals so displayed totals reconcile with their line items.
- Integrate M-Pesa mobile money payments with escrow release against approved project milestones, and build the interfaces that break a quoted amount into provider payout, platform commission and tax; also integrated S3 presigned-URL uploads, Firebase push notifications and Socket.IO real-time updates.
- Write and maintain automated tests with Vitest, Jest and the Node test runner, including integration and smoke suites and API contract-parity checks, reducing regression defects by approximately 30%.
- Investigate production issues systematically to root cause using browser DevTools, network inspection and API logs, typically closing 2-3 defects per sprint within agreed turnaround.
- Collaborate through Git feature-branch and pull request workflows, peer code review and Agile ceremonies, and built and delivered the end-to-end product demonstration used at the platform public launch.

PROJECTS

Real Estate Rental Listing Platform | React, Node.js, Express, PostgreSQL, Jest | 2023

- Built a rental listing platform with search and listing management: designed REST API endpoints and backend logic, applied relational database design and developed unit tests.

EDUCATION AND CERTIFICATIONS

Bachelor of Science, Information Technology | The Co-operative University of Kenya | 2019 – 2023

Relevant coursework: Web Development, Data Structures, Algorithms, OOP, Database Systems

Certifications: The React Crash Course (Pirple) | Node.js Masterclass, Backend | Frontend Fundamentals (freeCodeCamp)